The documentation and process conversion measures necessary to comply with this revision shall be completed by 16 December 1993

INCH-POUND

MIL-S-19500/317F
16 September 1993
SUPERSEDING
MIL-S-19500/317E
2 September 1980

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPE 2N2369A, 2N3227, AND 2N4449, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon high-speed switching transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3.

1.3 Maximum ratings.

Type	P_T T_A = +25°C		P_T T_C = +25°C		V_{CBO}	V_{EBO}	V_{CEO}	V_{CES}	T_{OP} and T_{STG}
	W		W		V dc	V dc	V dc	V dc	°C
2N2369A	0.36	1/	1.2	2/	40	4.5	15	40	-65
2N3227	0.36	1/	1.2	2/	40	6.0	20	40	to
2N4449	0.30	3/	1.5	4/	40	4.5	15	40	+200

1/ Derate linearly 2.06 mW/°C above T_A = +25°C.
2/ Derate linearly 6.85 mW/°C above T_C = +25°C.
3/ Derate linearly 1.71 mW/°C above T_A = +25°C.
4/ Derate linearly 8.56 mW/°C above T_C = +25°C.

1.4 Primary electrical characteristics.

Type	h_{FE2} 1/ V_{CE}= 0.4 V dc I_C = 30 mA dc		h_{FE4} 1/ V_{CE}= 1.0 V dc I_C = 100 mA dc		$R_{\theta JC}$ °C/mW	
	Min	Max	Min	Max	Min	Max
2N2369A	30	120	20	120		146
2N4449	30	120	20	120		117
2N3227	40	250	30	150		146

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Projects Office (NPPO), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Limits	$\lvert h_{FE} \rvert$ V_{CE} = 10 V dc I_C = 10 mA dc f = 100 MHz	$V_{CE(sat)1}$ I_C = 10 mA dc I_B = 1 mA dc	t_{on} I_C = 10 mA dc I_{B1} = 3 mA dc I_{B2} = -1.5 mA dc	t_{off} I_C = 10 mA dc I_{B1} = 3 mA dc I_{B2} = -1.5 mA dc	t_s
		V dc	ns	ns	ns
Min	5.0				
Max	10	0.20	12	18	13

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, appendix F, figure 9, T-2A for 2N2369A and 2N3227, and T-2B for 2N4449.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquistion document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{CES} and h_{FE3}	Not applicable
11	I_{CES}; h_{FE3}; ΔI_{CES} = 100 percent of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = 15 percent of initial value.	I_{CES} and h_{FE3}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CES} = 100 percent of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = 15 percent of initial value.	Subgroup 2 of table I herein; ΔI_{CES} = 100 percent of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = 15 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements in 4.5 of MIL-STD-750:

2N2369A, 2N3227 - - - V_{CB} = 12 V dc; P_T = 360 mW.
2N4449 - - - - - - - V_{CB} = 12 V dc; P_T = 300 mW.

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection (JANS). Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

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4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 12 V dc, P_T = 360 mW for 2N2369A and 2N3227; V_{CB} = 12 V dc, P_T = 300 mW for 2N4449; T_A = +25°C ±3°C for all devices; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 12 V dc, T_A = +125°C ±25°C for 96 hours; P_T = 360 mW for 2N2369A and 2N3227, P_T = 300 mW for 2N4449 at T_A = +100°C or adjusted as required by the chosen T_A to give an average T_J = +275°C.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	V_{CB} = 12 V dc, P_T = 360 mW for 2N2369A and 2N3227; V_{CB} = 12 V dc, P_T = 300 mW for 2N4449 at T_A = +30°C ±5°C. No heat sink or forced-air cooling on the devices shall be permitted.

4.4.4 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition
C2	2036	Test condition E.
C5	1026	V_{CB} = 12 V dc; P_T = 360 mW for 2N2369A and 2N3227; V_{CB} = 12 V dc, P_T = 300 mW for 2N4449 at T_A = +30°C ±5°C. No heat sink or forced-air cooling on devices shall be permitted.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage collector - emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2369A, 2N4449				15		V dc
2N3227				20		V dc
Breakdown voltage collector - emitter	3011	Bias condition C; I_C = 10 µA dc; V_{BE} = 0 V dc	$V_{(BR)CES}$	40		V dc
Breakdown voltage collector - base	3001	Bias condition D; I_C = 10 µA dc;	$V_{(BR)CBO}$	40		V dc
Breakdown voltage emitter - base	3026	Bias condition D; I_E = 10 µA dc;	$V_{(BR)EBO}$			
2N2369A, 2N4449				4.5		V dc
2N3227				6.0		V dc
Collector - emitter cutoff current	3041	Bias condition C; V_{CE} = 20 V dc	I_{CES}		0.4	µA dc
Collector - emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc; V_{BE} = 0.25 V dc	I_{CEX1}		0.3	µA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO}		0.25	µA dc
Collector - base cutoff current	3036	Bias condition D; V_{CB} = 20 V dc	I_{CBO1}		0.2	µA dc
Forward-current transfer ratio	3076	V_{CE} = 0.35 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE1}			
2N2369A, 2N4449				40	120	
2N3227				70	250	
Forward-current transfer ratio	3076	V_{CE} = 0.4 V dc; I_C = 30 mA dc; pulsed (see 4.5.1)	h_{FE2}			
2N2369A, 2N4449				30	120	
2N3227				40	250	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE3}			
2N2369A, 2N4449				40	120	
2N3227				100	300	
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 100 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N2369A, 2N4449				20	120	
2N3227				30	150	
Collector - emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{CE(sat)1}$		0.20	V dc
Collector - emitter voltage (saturated)	3071	I_C = 30 mA dc; I_B = 3.0 mA dc	$V_{CE(sat)2}$		.25	V dc
Collector - emitter voltage (saturated)	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		0.45	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)1}$	.70	0.85	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 30 mA dc; I_B = 3.0 mA dc	$V_{BE(sat)2}$		0.90	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)3}$	0.80	1.20	V dc
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector - base cutoff current	3036	Bias condition D; V_{CB} = 20 V dc	I_{CBO2}		30	µA dc
High-temperature operation:		T_A = +125°C				
Collector - emitter cutoff current	3041	Bias condition A; V_{CE} = 10 V dc; V_{BE} = 0.25 V dc	I_{CEX2}		30	µA dc
Collector - emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{CE(sat)4}$		0.3	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)4}$			
2N2369A, 2N4449				0.59		V dc
2N3227				0.50		V dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE5}			
2N2369A, 2N4449				20		
2N3227				40		
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)5}$		1.02	V dc
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 10 mA dc; f = 100 MHz	$\|h_{fe}\|$	5.0	10	
Open circuit output capacitance	3236	V_{CB} = 5 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		4.0	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}			
2N2369A, 2N4449					5.0	pF
2N3227					4.0	pF
Charge storage time		I_C = 10 mA dc; I_{B1} = 10 mA dc; I_{B2} = 10 mA dc; (see figure 1)	t_s			
2N2369A, 2N4449					13	ns
2N3227					18	ns
Turn-on time		I_C = 10 mA dc; I_{B1} = 3.0 mA dc; I_{B2} = 1.5 mA dc; (see figure 2)	t_{on}		12	ns
Turn-off time		I_C = 10 mA dc; I_{B1} = 3.0 mA dc; I_{B2} = 1.5 mA dc; (see figure 2)	t_{off}			
2N2369A, 2N4449					18	ns
2N3227					25	ns

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Groups B and C electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector - emitter cutoff current	3041	Bias condition C; V_{CE} = 20 V dc	I_{CES1}		.4	µA dc
2.	Collector - emitter cutoff current	3041	Bias condition C; V_{CE} = 20 V dc	I_{CES1}		.8	µA dc
3.	Emitter - base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO}		.25	µA dc
4.	Collector - emitter and resistance	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{CE(sat)1}$		.20	V dc
5.	Base - emitter voltage (saturated)	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)}$	0.70	0.85	V dc
6.	Forward-current transfer ratio	3076	V_{CE} = 0.35 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE1}			
	2N2369A, 2N4449 2N3227				40 70	120 250	
7.	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE3}			
	2N2369A, 2N4449 2N3227				40 100	120 300	
8.	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc I_C = 10 mA dc; pulsed (see 4.5.1)	Δh_{FE3}	±25 percent change from initial value.		
9.	Collector - emitter and resistance	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$\Delta V_{CE(sat)}$	±50 mV dc change from previous measured value.		
10.	Collector - emitter cutoff current	3041	Bias condition C; V_{CE} = 20 V dc;	ΔI_{CES1}	100 percent of initial value or 25 nA dc, whichever is greater.		

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 3, 4, 5, 6, and 7.

b. Subgroup 4, see table II herein, steps 1, 3, 5, 6, 7, and 9.

c. Subgroup 5, see table II herein, steps 1, 3, 4, 5, 6, 7, 9, and 10.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 6.

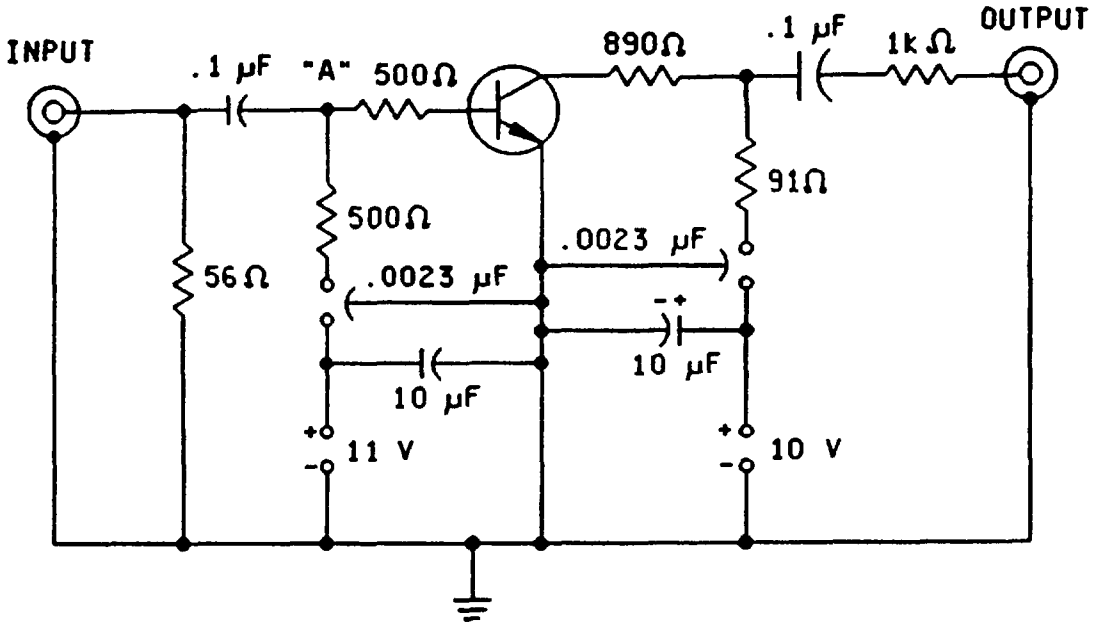
b. Subgroup 3 and 6, see table II herein, steps 2, 7, and 8.

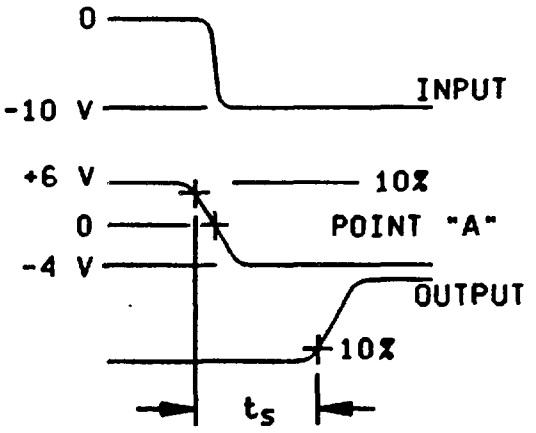
3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, table II herein, steps 1, 3, 4, 5, 6, and 7 (for JANS) and 1 and 6 (for JANTX and JANTXV).

b. Subgroup 6, see table II herein, steps 1, 3, 4, 5, 6, 7, 8, and 10 (for JANS) and 2, 7, and 8 (for JANTX and JANTXV).

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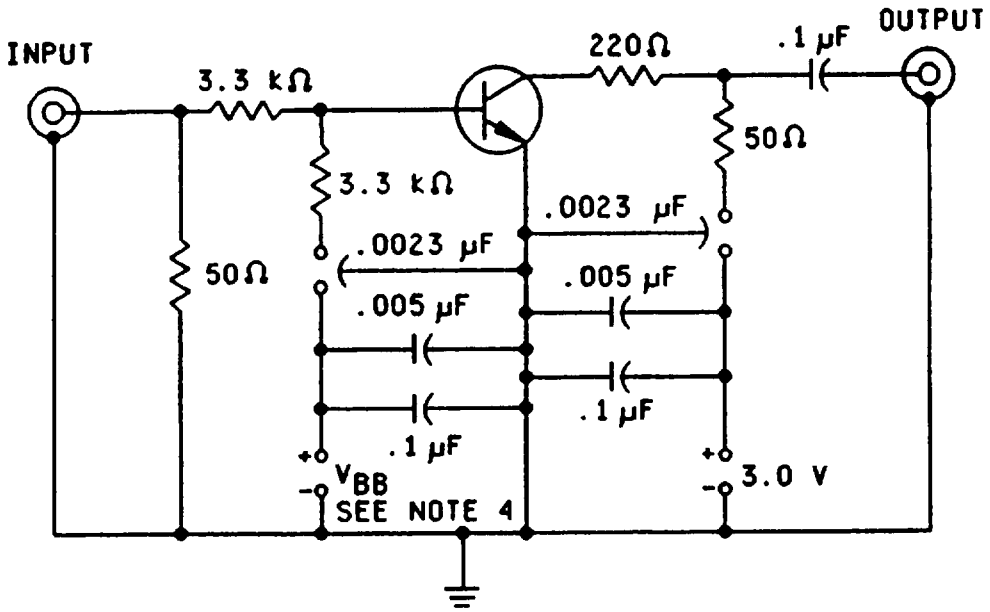


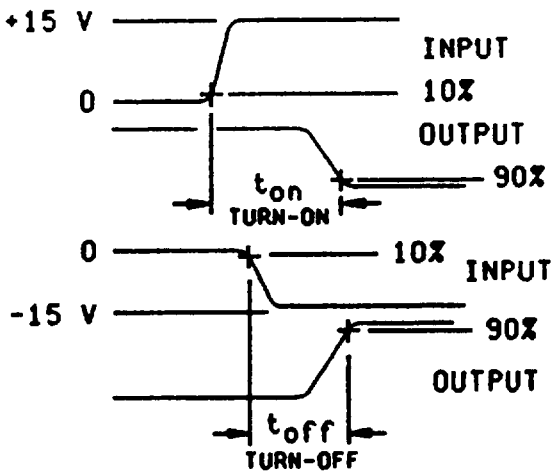
VOLTAGE WAVEFORMS

NOTES:
1. All capacitance in µF.
2. The input waveforms for each circuit are supplied by a pulse generator with the following characteristics: Z_{OUT} = 50Ω, t_r ≤ 1 ns, PW ≥ 300 ns, duty cycle ≤ 2 percent.
3. Output waveforms are monitored on an oscilloscope with the following characteristics: t_r ≤ .1 ns, Z_{IN} = 50Ω.

FIGURE 1. Charge storage time test circuit.

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VOLTAGE WAVEFORMS

NOTES:
1. All capacitance in μF.
2. The input waveforms for each circuit are supplied by a pulse generator with the following characteristics: Z_{OUT} = 50Ω, $t_r \leq 1$ ns, PW ≥ 300 ns, duty cycle ≤ 2 percent.
3. Input and output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq .1$ ns, Z_{IN} = 50Ω.
4. V_{BB} = -3.0 V for t_{on}, +12.0 V for t_{off}.

FIGURE 2. Turn-on and turn-off time test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/317F	2. DOCUMENT DATE (YYMMDD) 93/09/16

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING
TYPE 2N239A, 2N3227, AND 2N4449, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Gregory Rose	b. TELEPHONE (Include Area Code) (1) Commercial 301-731-8694	(2) AUTOVON
c. ADDRESS (Include Zip Code) NASA/Parts Projects Office (NPPO) NASA Goddard Space Flight Center Code 310.A Greenbelt, MD 20771	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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